

Research Memo - Best Optimizer Selection

Dasey Dang

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To: DAGE & Adam Ross Nelson

Research Question: [When using open ended descriptions to predict dominant transportation mode, which optimizer produces the best F1 score?]

Answer, Response, + Summary of Results

Adam is the most effective of the three optimizers tested for this neural-network architecture and data split. Using the transportation modeling data, I compared Adam, RMSprop, and stochastic gradient descent (SGD) while holding the neural-network architecture constant. The outcome included three classes: public transit dominant, rideshare dominant, and no dominant mode. The data were divided using a stratified 70/15/15 train, validation, and test split with a fixed seed of 1738. Because the outcome classes were imbalanced, macro-F1 was used as the primary validation metric. Adam produced the strongest validation performance with a macro-F1 of **0.4270** and validation accuracy of **0.6250**. RMSprop and SGD each produced a macro-F1 of **0.2342** and validation accuracy of **0.5417**.

Optimizer	Validation Macro-F1	Validation Accuracy
Adam	0.4270	0.6250
RMSprop	0.2342	0.5417
SGD	0.2342	0.5417

What questions or uncertainties remain?

The optimizer comparison was based on one data split and one set of learning-rate settings. Additional testing is needed to determine whether Adam remains the strongest optimizer across different splits, tuned learning rates, and class-weighting approaches. Would Adam remain the strongest optimizer if learning rates were separately tuned for Adam, RMSprop, and SGD?